P-S-320C
March 16, 1973
SUPERSEDING
Fed. Spec. P-S-320B
December 1, 1965

FEDERAL SPECIFICATION

SCOURING POWDER (GLASS CLEANING)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal Agencies.

1. SCOPE

1.1 This specification covers one type and grade of scouring powder for cleaning highly polished glass surfaces which must transmit light efficiently (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

SS-A-666	-	Asphalt, Petroleum (Built-Up Roofing, Water-Proofing and Damp-Proofing).
TT-P-350	-	Pigment, Lampblack-Dry.
PPP-B-636	-	Boxes, Shipping, Fiberboard.
PPP-C-55	-	Cans, Fiberboard and Paperboard (With Paper and Metal Ends).

Federal Standards:

FED-STD-123	-	Marking for Domestic Shipment (Civil Agencies).
FED-STD-536	-	Soap and Soap-Products (Including Synthetic Detergents); Sampling and Testing.

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards, as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in

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the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.

(Single copies of this specification and other Federal specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Services Centers at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles and Seattle, Washington.

(Federal Government activities may obtain copies of Federal Specifications and Standards, and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-L-6082 - Lubricating Oil; Aircraft Reciprocating (Piston) Engine.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Laws and Regulations:

21 CFR-191 - **Federal Hazardous Substances Act.**

(The code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for the issuance thereof.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

The National Formulary (N. F.).
The Pharmacopoeia of the United States (U. S. P.).

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(Application for copies should be addressed to the Mack Printing Company, Easton, Pa.)

The Soap and Detergent Association
Test Procedure and Standards - ABS and LAS Biodegradability
Scientific and Technical Report No. 3

(Application for copies should be addressed to the Soap and Detergent Association, 475 Park Avenue South, New York, New York 10016.)

(Copies for Military Activities are available from the Commanding Officer, Navy Publications and Forms Center, (Code 1032), 5801 Tabor Avenue, Philadelphia, Pa. 19120).

National Motor Freight Traffic Association, Inc., Agent

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, D.C. 20036.)

Uniform Classification Committee, Agent

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, Illinois 60606.)

Technical society and technical association specifications and standards are generally available for references from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 Materials. The scouring powder shall be a mixture of detergents, alkaline salts, and abrasives.

3.2 Composition. The scouring powder shall conform to the requirements of table I when tested as specified in 4.3.2.

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TABLE I. Composition

Characteristic	Percent by weight	
	Minimum	Maximum
Moisture and volatile matter	--	4.0
Total alkalinity of alcohol-insoluble matter calculated as sodium carbonate (Na_2CO_3)	--	5.0
Free alkali, calculated as sodium hydroxide (NaOH)	--	0.1
Anhydrous salt-free synthetic detergent (soap is excluded)	4.0	--
Matter insoluble in water	85.0	93.0
Matter insoluble in water retained on a No. 100 U.S. Standard Sieve	--	None
Matter insoluble in water retained on a No. 200 U.S. Standard Sieve	--	1.0
Rosin	None	None
Sugar	None	None
Biodegradability	90.0	--

3.3 Soil removal. The scouring powder shall remove all soil, leaving no scratches or streaks when tested as specified in 4.3.2.

3.4 Labeling. Each interior package shall have the following label clearly and legibly lithographed or stenciled on its face:

"SCOURING POWDER (GLASS CLEANING)

P-S-320

14 ounces"

Stock number

"DIRECTIONS FOR USE

Apply the scouring powder to a clean dampened cheesecloth and rub onto the soiled glass surface with a circular motion. Let it dry and then wipe off the dried film with a clean, soft, dry cloth."

Manufacturer's name
Manufacturer's address
Contract number
Date

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In addition, a proper precautionary label shall be furnished by the supplier in accordance with the Federal Hazardous Substances Act.

3.5 Workmanship. The scouring powder shall be uniform, homogeneous, white or light gray in color, clean, free from lumps and foreign matter. The powder shall not have a rancid or fishy odor.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.2.1 Inspection of materials and components. In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications and standards unless otherwise excluded, amended or qualified in this specification or applicable purchase document.

4.3 Inspection of the end item.

4.3.1 Examination of the end item. The end item shall be examined for the defects in the applicable subparagraphs at the inspection levels and acceptable quality levels (AQLs) set forth in 4.3.1.4. The lot size, for purpose of determining the sample size in accordance with MIL-STD-105, shall be expressed in number of filled unit containers of scouring powder for the examinations in 4.3.1.1 and 4.3.1.2, and in units of shipping containers for examinations in 4.3.1.3.

4.3.1.1 Examination of the end item for defects in appearance, form, color, and odor. The sample unit for this examination shall be one filled unit container of scouring powder.

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Examine	Defect
Appearance	Not clean; foreign matter present.
	Not uniform.
	Not homogeneous.
Form	Not a powder.
	Lump.
Color	Not white or light gray.
Odor	Rancid or fishy

4.3.1.2 Examination of end item for net weight. The sample unit shall be one filled unit container. Weigh each filled container separately, deducting the known tare weight to obtain the net weight per container. For lot acceptability, the average net weight per container shall be not less than the weight specified in the contract or order.

4.3.1.3 Examination of preparation for delivery. An examination shall be made to determine whether the packaging, packing and marking comply with the requirements of section 5. Defects shall be scored as specified in MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 with an AQL of 4.0, expressed in terms of defects per hundred units.

Examine	Defect
Containers	Not as specified
Contents	Not as specified
Markings	Omitted; incorrect; illegible; improper size, location, sequence or method of application.
Materials	Component missing or damaged
Workmanship	Bulging or distortion of containers. Cushioning inadequate, improper or missing.

4.3.1.3.1 Examination of closure and reinforcement of containers. When shipping containers are required to comply with PPP-B-636, examination for defects in closure and reinforcement shall be in accordance with the appendix of that specification.

4.3.1.4 Inspection levels and AQLs for examination. The inspection levels and the AQLs expressed in defects per 100 units, shall be as follows:

Examination paragraph	Inspection level	AQL
4.3.1.1	S-2	2.5
4.3.1.2	S-2	--

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4.3.2 Testing of the end item. The methods of testing specified in FED-STD-536 and listed in table II shall be followed for each lot. The lot size shall be expressed in unit containers of scouring powder. The sample unit for testing shall be the contents of four unit containers of scouring powder, selected from the lot. The sample size shall be as specified below. When specified in table II, tests shall be performed on a composite sample or on a sample unit basis. Two unit containers from each sample unit shall be thoroughly mixed to obtain the composite sample. Each sample unit and the composite shall be placed in separate, sealed, airtight, and watertight metal or glass containers. Two specimens of each sample unit, for moisture and volatile matter determination, shall be weighed immediately upon opening the sealed container. Care shall be exercised to prevent loss or absorption of moisture or loss of volatile matter during the sampling, weighing, and storage of the specimens and samples. The lot shall be unacceptable if a sample unit or the composite sample fails to meet any test requirement specified. All test reports shall contain the individual values utilized in expressing the final result.

Lot size (unit containers)	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 or more	5

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TABLE II. Instructions for testing of the end item

Characteristic	Specification reference		Requirements applicable to		Number determinations per unit	Results reported as
	Requirement	Test method	Sample unit	Comp. sample		
Moisture and volatile matter	Table I	5201 1/	X		2	Average of 2 determinations to nearest 0.1 percent.
Alkaline salts	Table I	5901 1/		X	2	Average of 2 determinations to nearest 0.1 percent.
Free alkali	Table I	5301 1/		X	2	Average of 2 determinations to nearest 0.1 percent.
Synthetic organic detergent	Table I	6501 1/		X	2	Average of 2 determinations to nearest 0.1 percent.
Matter insoluble in water	Table I	5801 1/		X	2	Average of 2 determinations to nearest 0.1 percent.
Sieve tests:						
No. 100	Table I	2101 1/		X	1	Pass or fail.
No. 200	Table I	2101 1/		X	1	Pass or fail.
Rosin (qualitative test)	Table I	6301 1/	X		1	Pass or fail.
Sugar (qualitative test)	Table I	1701 1/	X		1	Pass or fail.
Performance	3.3	4.4.2	X		1	Pass or fail.
Biodegradability	Table I	4.4.3		X	2	Average of 2 determinations to nearest 0.1 percent.

1/ FED-STD-536

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4.4 Test procedures.

4.4.1 Soiling mixture. Mix the ingredients in table III to form a smooth paste of uniform composition for use as a soiling mixture.

TABLE III. Composition of soiling mixture

Ingredient	Parts by weight
Wool fat, U.S.P.	50.0
Asphalt (conforming to SS-A-666, type II)	15.0
Turpentine oil, N.F.	25.0
Lubricating oil (conforming to MIL-L-6082, grade 1100)	20.0
Lampblack (conforming to TT-P-350)	20.0

4.4.2 Performance. Apply a thin even film of the soiling mixture (see 4.4.1) to one side of a clean, scratch-free, 4- by 4-inch window-glass panel with a clean, dry cloth. Weigh out 5± 0.001 g of scouring powder, spread on a 4- by 4-inch wad of clean dampened cheesecloth, and apply to the soiled glass panel with a circular hand motion. Let it dry forming a film. Wipe the dried film off the glass panel with another 4- by 4-inch clean dry cloth. Note cleanliness, also if any scratches, soil residue, or streaks.

4.4.3 Biodegradability of the synthetic detergent in the scouring powder. The supplier shall submit a certificate of compliance to the requirements for biodegradability. The certificate shall be accompanied by actual test data (field or laboratory), including the detail test procedure utilized in making the biodegradability determination. When the detergent is an alkyl benzene sulfonate or a linear alklate sulfonate, the biodegradability shall be determined in accordance with the Test Procedure and Standards - ABS and LAS Biodegradability, Scientific and Technical Report No. 3 of The Soap and Detergent Association.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B, or C as specified (see 6.2).

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5.1.1 Level A. Fourteen ounces of scouring powder shall be packaged in a can conforming to type V, class 2 of PPP-C-55.

5.1.2 Level B (Civil Agencies). Fourteen ounces of scouring powder shall be packaged in a commercial type, spirally or convolutely wound fiberboard or paperboard can. The commercial type cans shall be fitted with metal or plastic ends with one of the ends provided with not less than six punch out holes or prepunched holes covered with pressure sensitive tape.

5.1.3 Level C. Scouring powder shall be packaged to afford adequate protection against physical damage and deterioration during shipment from the supply source to the first receiving activity. The supplier may use his standard practice when it meets these requirements.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. Forty-eight cans of scouring powder, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC-L, grade V2s of PPP-B-636. The inside of each fiberboard container shall be fitted with a taped box liner conforming to type CF, class weather-resistant, variety DW, grade V15c of PPP-B-636. Tiers of cans (when applicable) shall be separated by fiberboard pads fabricated of weather-resistant grade material. Each shipping container shall be closed in accordance with method III, waterproofed in accordance with method V, and reinforced as specified in the appendix of the container specification.

5.2.2 Level B. Forty-eight cans of scouring powder, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. Tiers of cans (when applicable) shall be separated by fiberboard pads fabricated of the same material as the shipping container. Each shipping container shall be closed in accordance with method II as specified in the appendix of the container specification.

5.2.2.1 When specified (see 6.2), the fiberboard shipping container shall be a grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with method III as specified in the appendix of the container specification.

5.2.3 Level C. Scouring powder, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

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5.3 Marking.

5.3.1 Civil Agencies. In addition to the labeling specified in 3.4, and to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with FED-STD-123.

5.3.2 Military requirements. In addition to the labeling specified in 3.4, and to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Scouring powder covered by this specification is intended for safe cleaning of glass or other vitreous surfaces.

6.2 Ordering data. Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Selection of the applicable levels of packaging and packing (see 5.1 and 5.2).
(c) When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).

MILITARY CUSTODIANS

Army - GL
Air Force - 84

Review activity:

Army - MD

User activities:

Navy - YD, MC

Preparing activity:

Army - GL

Civil Agency Interest:

GSA-FSS
HEW-NIH

Project No. 7930-0279

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

Scouring Powder (Glass Cleaning) P-S-320C

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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